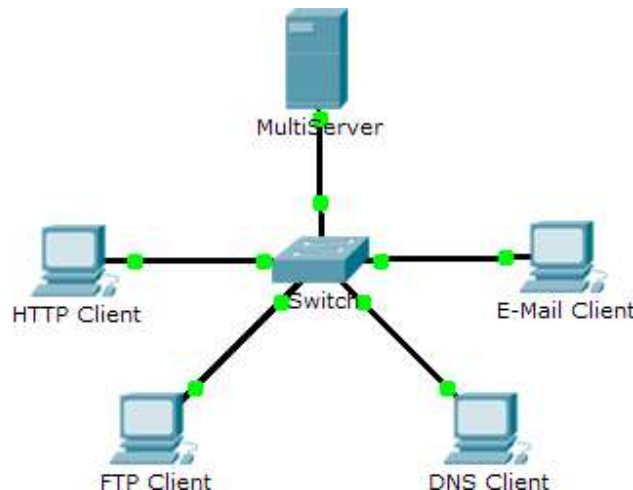


Packet Tracer Simulation - TCP and UDP Communications

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Topology



Objectives

Part 1: Generate Network Traffic in Simulation Mode

Part 2: Examine the Functionality of the TCP and UDP Protocols

Background

This simulation activity is intended to provide a foundation for understanding the TCP and UDP in detail. Simulation mode provides the ability to view the functionality of the different protocols.

As data moves through the network, it is broken down into smaller pieces and identified in some fashion so that the pieces can be put back together. Each of these pieces is assigned a specific name (protocol data unit [PDU]) and associated with a specific layer. Packet Tracer Simulation mode enables the user to view each of the protocols and the associated PDU. The steps outlined below lead the user through the process of requesting services using various applications available on a client PC.

This activity provides an opportunity to explore the functionality of the TCP and UDP protocols, multiplexing and the function of port numbers in determining which local application requested the data or is sending the data.

Part 1: Generate Network Traffic in Simulation Mode

Step 1: Generate traffic to populate Address Resolution Protocol (ARP) tables.

Perform the following tasks to reduce the amount of network traffic viewed in the simulation.

- Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.
- Enter the **ping 192.168.1.255** command. This will take a few seconds as every device on the network responds to **MultiServer**.
- Close the **MultiServer** window.

Step 2: Generate web (HTTP) traffic.

- Switch to Simulation mode.
- Click **HTTP Client** and click the **Desktop** tab > **Web Browser**.
- In the URL field, enter **192.168.1.254** and click **Go**. Envelopes (PDUs) will appear in the simulation window.
- Minimize, but do not close, the **HTTP Client** configuration window.

Step 3: Generate FTP traffic.

- Click **FTP Client** and click the **Desktop** tab > **Command Prompt**.
- Enter the **ftp 192.168.1.254** command. PDUs will appear in the simulation window.
- Minimize, but do not close, the **FTP Client** configuration window.

Step 4: Generate DNS traffic.

- Click **DNS Client** and click the **Desktop** tab > **Command Prompt**.
- Enter the **nslookup multiserver.pt.ptu** command. A PDU will appear in the simulation window.
- Minimize, but do not close, the **DNS Client** configuration window.

Step 5: Generate Email traffic.

- Click **E-Mail Client** and click the **Desktop** tab > **E Mail** tool.
- Click **Compose** and enter the following information:
 - To:** user@multiserver.pt.ptu
 - Subject:** Personalize the subject line
 - E-Mail Body:** Personalize the Email
- Click **Send**.
- Minimize, but do not close, the **E-Mail Client** configuration window.

Step 6: Verify that the traffic is generated and ready for simulation.

Every client computer should have PDUs listed in the Simulation Panel.

Part 2: Examine Functionality of the TCP and UDP Protocols

Step 1: Examine multiplexing as all of the traffic crosses the network.

You will now use the **Capture/Forward** button and the **Back** button in the Simulation Panel.

- Click **Capture/Forward** once. All of the PDUs are transferred to the switch.
- Click **Capture/Forward** again. Some of the PDUs disappear. What do you think happened to them?
[Some PDUs disappear because they are delivered to their destinations. This represents data being forwarded through network and successfully received by the appropriate device.](#)
- Click **Capture/Forward** six times. All clients should have received a reply. Note that only one PDU can cross a wire in each direction at any given time. What is this called?

[Half-duplex communication](#)

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- d. A variety of PDUs appears in the event list in the upper right pane of the simulation window. Why are they so many different colors?

Each color represents a specific protocol or type of communication being used. This allows users to easily differentiate and track the flow of different PDUs through the network simulation.

- e. Click **Back** eight times. This should reset the simulation.

Note: Do not click **Reset Simulation** any time during this activity; if you do, you will need to repeat the steps in Part 1.

Step 2: Examine HTTP traffic as the clients communicate with the server.

- a. Filter the traffic that is currently displayed to display only **HTTP** and **TCP** PDUs filter the traffic that is currently displayed:
- 1) Click **Edit Filters** and toggle the **Show All/None** check box.
 - 2) Select **HTTP** and **TCP**. Click anywhere outside of the Edit Filters box to hide it. The Visible Events should now display only **HTTP** and **TCP** PDUs.
- b. Click **Capture/Forward**. Hold your mouse above each PDU until you find one that originates from **HTTP Client**. Click the PDU envelope to open it.
- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?
TCP.

Are these communications considered to be reliable?

Yes.

- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

SRC PORT: 1027
DEST PORT: 80
SEQUENCE NUM: 0
ACK NUM: 0

Left of the WINDOW field:

OFFSET: 0x0
RESERVED: 0
FLAGS: 0b00000010

- e. Close the PDU and click **Capture/Forward** until a PDU returns to the **HTTP Client** with a checkmark.
- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

SRC PORT: 80

SEQUENCE NUMBER: 0

The source port changes from 1027 to 80 because the communication direction switches from client-to-server to server-to-client. The sequence number has no difference than before.

- g. There is a second **PDU** of a different color, which **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the HTTP communication. Click this second PDU envelope and select **Outbound PDU Details**.

- h. What information is now listed in the TCP section? How are the port and sequence numbers different

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from the previous two PDUs?

SRC PORT: 1027

SEQUENCE NUMBER: 1

The sequence number change to 1 indicating the start of packet transmission. The sequence number is incrementing to track the new communication phase.

- i. Click **Back** until the simulation is reset.

Step 3: Examine FTP traffic as the clients communicate with the server.

- a. In the Simulation Panel, change **Edit Filters** to display only **FTP** and **TCP**.
- b. Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **FTP Client**. Click that PDU envelope to open it.
- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?

TCP.

Are these communications considered to be reliable?

Yes.

- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

SRC PORT: 1025

DEST PORT: 21

SEQUENCE NUM: 0

ACK NUM: 0

Left of the WINDOW field:

OFFSET: 0x0

RESERVED: 0

FLAGS: 0b00000010

- e. Close the PDU and click **Capture/Forward** until a PDU returns to the **FTP Client** with a checkmark.
- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

SRC PORT: 21

SEQUENCE NUMBERS: 0

The source port is change to 21 indicates the response is from FTP server which differ from the previous outbound PDU where the source put was 1025. Meanwhile, sequence number remains 0 showing no packet transfer has occurred yet.

- g. Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?

SRC PORT: 1025

SEQUENCE NUMBER: 1

The source port remains 1025 indicates the communication is still from the FTP client. However, the SEQUENCE NUMBER has incremented to 1 meaning the start of transmission of the packet transfer.

- h. Close the PDU and click **Capture/Forward** until a second PDU returns to the **FTP Client**. The PDU is a different color.
- i. Open the PDU and select **Inbound PDU Details**. Scroll down past the TCP section. What is the message from the server?

Welcome to PT Ftp server.
- j. Click **Back** until the simulation is reset.

Step 4: Examine DNS traffic as the clients communicate with the server.

- a. In the Simulation Panel, change **Edit Filters** to display only **DNS** and **UDP**.
- b. Click the PDU envelope to open it.
- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?
DNS Query.

Are these communications considered to be reliable?
No.

- d. Record the **SRC PORT** and **DEST PORT** values. Why is there no sequence and acknowledgement number?

SRC PORT: 1025

DEST PORT: 53

The communication uses UDP which is a connectionless protocol. UDP does not establish connection or maintain a session so it does not use sequence number and acknowledgement number for tracking and ensuring packet delivery.

- e. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **DNS Client** with a checkmark.
- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

SRC PORT: 53

DESTINATION NUMBER: 1025

The source port is now 53 indicating the response is from the DNS server. There is still no sequence number.

- g. What is the last section of the **PDU** called?
DNS Answer

- h. Click **Back** until the simulation is reset.

Step 5: Examine email traffic as the clients communicate with the server.

- a. In the Simulation Panel, change **Edit Filters** to display only **POP3**, **SMTP** and **TCP**.

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- b. Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **E-mail Client**. Click that PDU envelope to open it.
- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What transport layer protocol does email traffic use?

TCP.

Are these communications considered to be reliable?

Yes.

- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

SRC PORT: 1025

DEST PORT: 25

SEQUENCE NUM: 0

ACKNUM: 0

Left of the WINDOW field:

OFFSET: 0x0

RESERVED: 0

FLAGS: 0b00000010

- e. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **E-Mail Client** with a checkmark.
- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

SRC PORT: 25

SEQUENCE NUMBER: 0

The Source port is now 25 indicating the response is from the email server while the sequence number remains 0.

- g. Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?

SOURCE PORT: 1025

SEQUENCE NUMBER: 1

The source port remains 1025 indicates the communication is still from the email client. However, the sequence number has incremented to 1 indicating the progression of the packet transfer.

- h. There is a second **PDU** of a different color that **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the email communication. Click this second PDU envelope and select **Outbound PDU Details**.
- i. How are the port and sequence numbers different from the previous two PDUs?

SOURCE PORT: 1025

SEQUENCE NUMBER: 1

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The source port remains 1025 indicates the communication is from email client. However, the sequence number is now 1 indicates the start of packet transfer.

- j. What email protocol is associated with TCP port 25? What protocol is associated with TCP port 110?
TCP port 25: SMTP
TCP port 110: POP3
- k. Click **Back** until the simulation is reset.

Step 6: Examine the use of port numbers from the server.

- a. To see TCP active sessions, perform the following steps in quick succession:
- 1) Switch back to **Realtime** mode.
 - 2) Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.
- b. Enter the **netstat** command. What protocols are listed in the left column?

TCP.

What port numbers are being used by the server?

21

- c. What states are the sessions in?
ESTABLISHED
- d. Repeat the **netstat** command several times until you see only one session still ESTABLISHED. For which service is this connection still open?
FTP control session

Why doesn't this session close like the other three? (Hint: Check the minimized clients)

The FTP remains open to handle commands from client. The FTP control connection stays active until the FTP client is explicitly closed or terminated.

Suggested Scoring Rubric

Activity Section	Question Location	Possible Points	Earned Points
Part 2: Examine Functionality of the TCP and UDP Protocols	Step 1	15	
	Step 2	15	
	Step 3	15	
	Step 4	15	
	Step 5	15	
	Step 6	25	
Total Score		100	

